

# Notes on Infinite Products

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June 2026

## 1 Introduction

A standard infinite product can be generalized in the form given below:

$$p_N = \prod_{n=1}^{\infty} a_n, \quad \prod_{n=1}^{\infty} a_n = a_1 \cdot a_2 \cdot a_3 \cdot a_4 \cdots$$

If the value of the product  $p_N$  is not equal to zero, the infinite product is convergent and its value equals to L.

$$p_N = \prod_{n=1}^{\infty} a_n, \quad \boxed{\lim_{n \rightarrow \infty} p_N = L \neq 0}$$

## 2 Algebraic Properties

Infinite products have four different algebraic properties:

### 2.1 Product Property

If two infinite products are convergent, the product of those infinite products equals to the infinite product of the products of their general terms

$$\boxed{\prod_{n=k}^{\infty} a_n \cdot \prod_{n=k}^{\infty} b_n = \prod_{n=k}^{\infty} ab_n}$$

**Proof**

$$\begin{aligned} \prod_{n=k}^{\infty} a_n &= a_k \cdot a_{k+1} \cdot a_{k+2} \cdots, \quad \prod_{n=k}^{\infty} b_n = b_k \cdot b_{k+1} \cdot b_{k+2} \cdots \\ \prod_{n=k}^{\infty} a_n \cdot \prod_{n=k}^{\infty} b_n &= a_k \cdot b_k \cdot a_{k+1} \cdot b_{k+1} \cdot a_{k+2} \cdot b_{k+2} \cdots = ab_k \cdot ab_{k+1} \cdot ab_{k+2} \cdots \\ \therefore \prod_{n=k}^{\infty} a_n \cdot \prod_{n=k}^{\infty} b_n &= \prod_{n=k}^{\infty} ab_n \end{aligned}$$

## 2.2 Exponential Power Property

An infinite product to the power of  $m$  is equal to the infinite product of the  $m$ th power of its general term

$$\left[ \prod_{n=k}^{\infty} (a_n) \right]^m = \prod_{n=k}^{\infty} (a_n)^m$$

**Proof**

$$\begin{aligned} \prod_{n=k}^{\infty} a_n &= a_k \cdot a_{k+1} \cdot a_{k+2} \cdot \dots, \quad \left[ \prod_{n=k}^{\infty} a_n \right]^m = (a_k)^m \cdot (a_{k+1})^m \cdot (a_{k+2})^m \cdot \dots \\ &= (a_k \cdot a_{k+1} \cdot a_{k+2} \cdot \dots)^m = \prod_{n=k}^{\infty} (a_n)^m \end{aligned}$$

## 2.3 Quotient Property

If two infinite products are convergent, the quotient of those infinite products equals to the infinite product of the quotient of their general terms

$$\frac{\prod_{n=k}^{\infty} a_n}{\prod_{n=k}^{\infty} b_n} = \prod_{n=k}^{\infty} \frac{a_n}{b_n}$$

**Proof**

$$\begin{aligned} \prod_{n=k}^{\infty} a_n &= a_k \cdot a_{k+1} \cdot a_{k+2} \cdot \dots, \quad \prod_{n=k}^{\infty} b_n = b_k \cdot b_{k+1} \cdot b_{k+2} \cdot \dots \\ \frac{\prod_{n=k}^{\infty} a_n}{\prod_{n=k}^{\infty} b_n} &= \frac{a_k \cdot a_{k+1} \cdot a_{k+2} \cdot \dots}{b_k \cdot b_{k+1} \cdot b_{k+2} \cdot \dots} = \frac{a_k}{b_k} \cdot \frac{a_{k+1}}{b_{k+1}} \cdot \frac{a_{k+2}}{b_{k+2}} \cdot \dots = \prod_{n=k}^{\infty} \frac{a_n}{b_n} \end{aligned}$$

## 2.4 Logarithmic Linearity property

Logarithmic linearity property valid if and only if the infinite product converges to a nonzero value. This property allows us to easily analyze convergence or divergence of infinite products. Infinite products are additive under natural logarithm

$$\ln \left( \prod_{n=k}^{\infty} a_n \right) = \sum_{n=k}^{\infty} \ln(a_n)$$

### Proof

$$\prod_{n=k}^{\infty} a_n = a_k \cdot a_{k+1} \cdot a_{k+2} \cdot \dots, \quad \ln \left( \prod_{n=k}^{\infty} a_n \right) = \ln(a_k \cdot a_{k+1} \cdot a_{k+2} \cdot \dots)$$

$$\ln(p \cdot q) = \ln(p) + \ln(q) \quad \therefore \quad \ln(a_k \cdot a_{k+1} \cdot a_{k+2} \cdot \dots) = \ln(a_k) + \ln(a_{k+1}) + \ln(a_{k+2}) \cdot \dots$$

$$\ln(a_k) + \ln(a_{k+1}) + \ln(a_{k+2}) \cdot \dots = \sum_{n=k}^{\infty} \ln(a_n), \quad \therefore \quad \ln \left( \prod_{n=k}^{\infty} a_n \right) = \sum_{n=k}^{\infty} \ln(a_n)$$

## 2.5 A Different Case of Logarithmic Linearity

The property of logarithmic linearity is valid not only for the cases where the natural logarithm of the product is being taken, but also the cases where the logarithm of some other base of the product is being taken. Consider an infinite product whose logarithm of some base  $r$  is being taken.

$$\log_r \left( \prod_{n=k}^{\infty} a_n \right)$$

Since any base of logarithm can be expressed as a form of  $\ln$ , we can write:

$$\log_r(Q) = \frac{\ln(Q)}{\ln(r)} \quad (r > 1) \quad , \quad \log_r \left( \prod_{n=k}^{\infty} a_n \right) = \frac{1}{\ln(r)} \cdot \ln \left( \prod_{n=k}^{\infty} a_n \right)$$

$$\ln \left( \prod_{n=k}^{\infty} a_n \right) = \sum_{n=k}^{\infty} \ln(a_n) \quad \therefore \quad \boxed{\log_r \left( \prod_{n=k}^{\infty} a_n \right) = \frac{1}{\ln r} \cdot \sum_{n=k}^{\infty} \ln(a_n)}$$